Project of Data Visualization: Milestone 2

Group *EUFlow*: Amine Youssef, Lia Lee, Hsiangtien Kuo

1. Project Overview and Objectives

Our project aims to develop an interactive web-based visualization platform to explore tourism patterns across Europe. The goal is to help users understand how Europeans travel by visualizing key aspects of tourism behavior, including where people travel (destination popularity), when they travel (seasonality), and how much they spend (expenditure).

These dimensions will be presented through coordinated visualizations such as maps, heatmaps, and comparative charts, allowing users to explore patterns across countries and over time. The platform is designed to support both high-level exploration and more detailed analysis through interactive features. In addition to the European overview, we include a focused section on Switzerland, where we examine how geographic distance and economic factors (GDP) relate to tourism flows.

2. Project Implementation Plan

2.1. Minimum Viable Product (MVP)

The MVP of our project consists of a website structured into two main sections: tourism in Europe and tourism in Switzerland. The goal of the MVP is to implement a set of core visualizations that allow users to explore the main aspects of tourism patterns in a clear and intuitive way.

- **Page 1: Tourism in Europe**
  - Destination popularity
    - **Choropleth map** showing the number of visitors per country
    - **Ranking bar chart** highlighting the top 10 destinations
    - **Heatmap** displaying how destination popularity evolves over time
  - Seasonality
    - **Choropleth map** that captures seasonal differences across countries
  - Travel expenditure
    - **Bar chart** showing the top and bottom destinations in terms of travel spending
- **Page 2: Tourism in Switzerland**
  - Destination popularity within Switzerland
    - **Choropleth map** at the canton level to visualize the distribution of tourism
  - Factors influencing tourism to Switzerland
    - **Scatter plot** (distance vs. visitors) to analyze geographic proximity effects
    - **Scatter plot** (GDP per capita vs. visitors) to analyze economic factors

2.2. Extra Ideas

While the MVP focuses on core visualizations to convey key tourism patterns, we aim to further enhance the platform by introducing additional interactive features, animations, and more advanced visual explorations. These enhancements are designed to support more intuitive data exploration, allowing users not only to observe patterns but also to actively engage with the data and uncover deeper insights. Although not essential for basic functionality, these elements significantly improve usability, user engagement, and the overall analytical experience.

- **Interactive Tooltips:** Display detailed information such as the number of visitors and expenditure when users hover over countries or regions.
- **Clickable Elements with Drill-down Functionality:** Allow users to click on countries to access more detailed views, for example from the European level to the map of Switzerland.
- **Time Slider with Animation:** Enable users to explore changes over time through a slider combined with animated transitions.
- **Filtering and Comparison:** Allow users to select specific countries, regions, or time periods in order to compare tourism patterns across different contexts.
- **Enhanced Map Interactions:** Support zooming and panning to provide more flexible and detailed exploration of the map.

2.3. Tools

For development of the high-fidelity prototype, we will use Figma to design the structure and layout of the website. For implementation, we plan to use D3.js as the main visualization library, since it provides the flexibility needed for interactive maps, heatmaps, bar charts, and scatter plots. The website will be developed based on the designs created in Figma, through Node.js, HTML, CSS, and JavaScript. We will use Python for data cleaning, preprocessing, and exploratory analysis, and GeoJSON/TopoJSON for geographic visualizations of maps.

2.4. Lectures

In further development of the website and visualizations of the data, we will build on several key lectures from the course. The project will be implemented with the content and concepts from Lectures: *JavaScript Part 1*, *More JavaScript*, and *Basic Web Development*. In addition, techniques from Lectures: *D3.js*, *More Interactive D3.js*, *Maps*, and *Practical Maps* will support the development of map-based visualizations and the geographic data. Furthermore, we will follow the principles outlined in *Do's and Don'ts in Visualization* to ensure effective and clear design of the visualization, and incorporate ideas from the *Storytelling* lecture to present the data in a coherent and engaging way.

3. Sketches

The following sketches of the visualization are designed using Figma.

To illustrate geographical patterns of tourism in Europe and Switzerland, choropleth maps are used to visualize spatial and seasonal variations in visitor numbers across European countries, as well as across Swiss cantons in the hotel sector (Fig. 1).

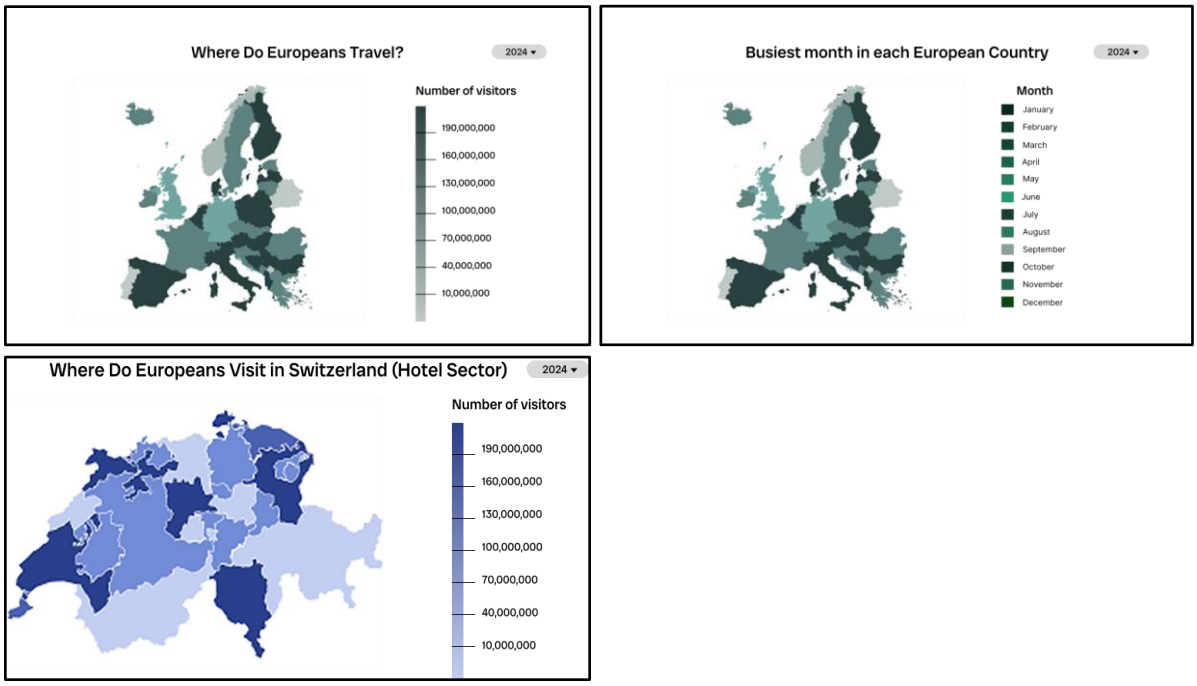

Figure 1: Choropleth Maps

The visualizations are designed using a bar chart (Fig. 2) and a stacked bar chart (Fig. 3) to represent the tourism data. The bar chart is used as a ranking chart to compare the top 10 most visited countries in Europe. To represent the percentage share of each country, stacked bar chart is used to effectively communicate proportions of a whole. This visualization illustrates the share of total tourism by each country, with a consistent color gradient applied to enhance visual hierarchy and readability.

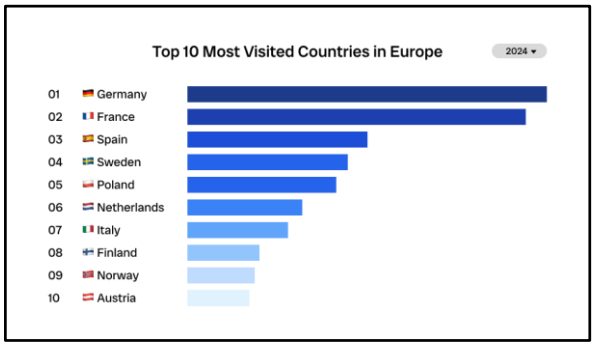

Figure 2: Bar Chart

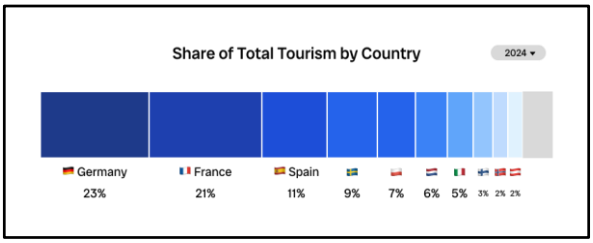

Figure 3: Stacked Bar Chart

To analyze the factors influencing tourism to Switzerland, we use line chart (Fig. 4) to show the average expenditure per night for the top five European countries with the highest spending, and scatter plots (Fig. 5) to examine the relationships between the distance of European countries from Switzerland and the number of visitors from European countries traveling to Switzerland, as well as GDP per capita and visitor numbers, in order to explore geographic and economic effects.

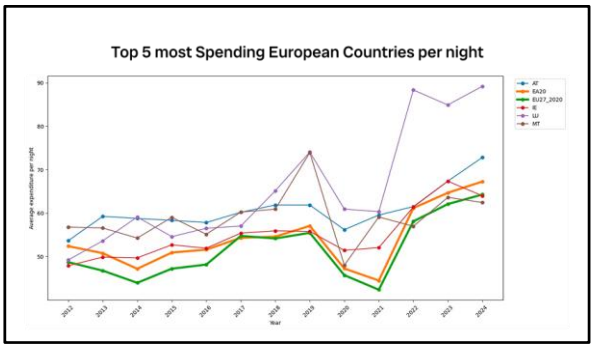

Figure 4: Line Chart

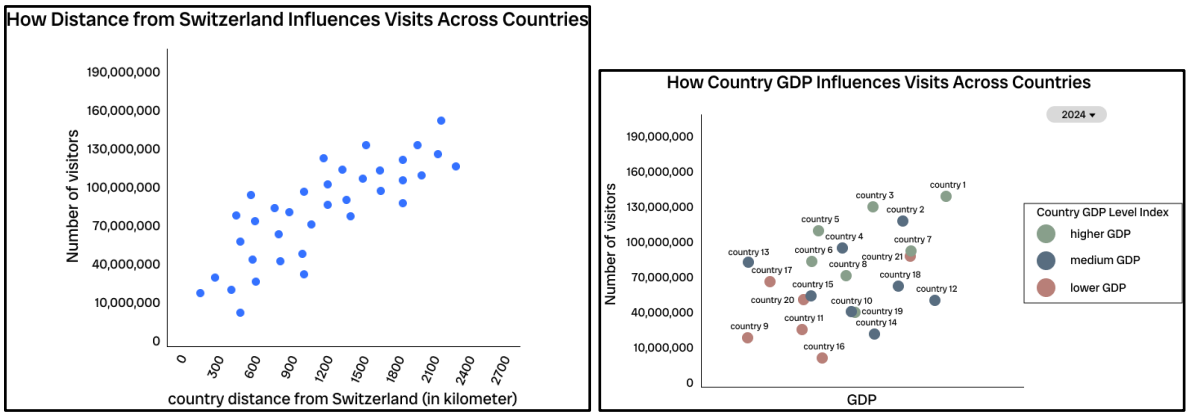

Figure 5: Scatter Plots